Project: comparing RL methods in simulated mean reverting stock

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Project Goal. Compare various RL methods across different data-driven settings, including on-policy versus off-policy approaches, and the use of a simulation oracle versus reliance solely on sequential data.

Price dynamics. Let S_t be the stock price with $S_0 = 1$ and $L_t := \log \frac{S_t}{S_{t-1}}$ be the return rate. For ease of implementation and debugging, we consider the 1-dimensional mean-reverting process for L:

$$L_{t+1} = L_t - \kappa L_t + \sigma Z_t, \quad L_0 = 0,$$

where $\kappa \in (0, 1)$, $\sigma > 0$ are model parameters and Z_t is an independent standard normal random variable (Brownian increment).

Objective. Let A_t be the number of shares invested in the asset at time t. Here we allow fractional share and A_t take valued in $\mathbb{R}$. We define the single-period reward as:[1]

$$R_t := A_t(S_{t+1} - S_t) - \lambda A_t^2, \tag{1}$$

eq:Reward1

where $\lambda > 0$ is a regularization parameter that penalizes extreme positions. Alternatively, a more realistic set up involves a stylized model for transaction cost:

$$R_t := A_t(S_{t+1} - S_t) - \lambda(A_t - A_{t-1})^2 S_t \tag{2}$$

eq:Reward2

with $A_{-1} = 0$ for $t = 0$. The objective is to find a trading policy π that maximizes the expected discounted total reward:

$$\max_{\pi} \ \mathbb{E}^{\pi}\left[\sum_{t=0}^{\infty} \gamma^t R_t \,\middle|\, S_0 = 1, L_0 = 0, A_{-1} = 0\right]$$

where $\gamma \in (0, 1)$ is the discount factor.

Methods. Employ RL methods to solve the optimization problem above. Execute your methods for Reward (1) and Reward (2) separately. For the simpler Reward (1), derive the optimal control policy analytically to serve as a benchmark. Compare the performance of RL algorithms under two distinct settings: (a) with access to a simulation oracle, and (b) with access only to sequential data.

Submission. Please submit a written report and a Jupyter notebook that together demonstrate your implementation. The written report should address the following points:

- Describe the methods for sampling trajectories from the controlled price dynamics, for both the simulation oracle and sequential data settings. If using off-policy methods, detail the policy used for data collection.

[1] The self-financing condition is implicitly assumed, allowing unlimited borrowing/lending from a risk-free money market account at zero interest rate for simplicity.

- Provide an algorithmic description of the RL methods used. Describe the function approximation architecture (e.g., neural networks) in detail, including the inputs (state representation) and outputs (action or value). Clearly clarify the update procedures for the policy and value function (if applicable).
- Test the performance across various sets of model parameters $(\kappa, \sigma, \lambda, \gamma)$. Evaluate the final policy using independently generated simulation paths. Report the mean discounted (and properly truncated) total reward estimated via Monte Carlo. Performance criteria beyond the primary objective (e.g., Sharpe ratio, maximum drawdown) are encouraged but not required.
- Visualize the learned policy. Compare the visualization across various model parameters. For Reward (1), compare the learned policy with the theoretically derived optimal policy.
- Conduct at least one ablation study on a key training hyper-parameter (e.g., termination condition, learning rate, network architecture).